VIRTUAL UNIVERSITY OF PAKISTAN

A Good Education is a Foundation For a Better Future

CORRECT ANSWER SOLVED BY HADI
EMAIL:hadirajputofficial@gmail.com
WHATSAPP& CALL:03087122922

(Final Term Past Paper)

MADE AND SOLVED BY TEAM HADI

WARNING: Team HADI is not responsible for any mistake or wrong answer. All students reading and using this document may check and confirm the answers at their own.

WhatsApp:03087122922

FACEBOOK ID: https://www.facebook.com/hadipastpapers/

Best of luck!

PAST PAPER

Question No : 1 of 52 **Marks: 1 (Budgeted Time 1 Min)**

Which one of the following is correct Wallis Cosine formula when n is odd and $n \geq 3$?

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Answer (Please select your correct option)

- $\int_0^{\frac{\pi}{2}} \cos^n x\, dx = \frac{\pi}{2} \frac{(n-1)}{n} \frac{(n-3)}{(n-2)} \frac{(n-5)}{(n-4)} \cdots\cdots\cdots \frac{5}{6} \frac{3}{4} \frac{1}{2}$
- $\int_0^{\frac{\pi}{2}} \cos^n x\, dx = \frac{\pi}{2} \frac{(n)}{(n-1)} \frac{(n-2)}{(n-3)} \frac{(n-4)}{(n-5)} \cdots\cdots\cdots \frac{6}{5} \frac{4}{3} \frac{2}{1}$
- $\int_0^{\frac{\pi}{2}} \cos^n x\, dx = \frac{(n)}{(n-1)} \frac{(n-2)}{(n-3)} \frac{(n-4)}{(n-5)} \cdots\cdots\cdots \frac{6}{5} \frac{4}{3} \frac{2}{1}$
- $\int_0^{\frac{\pi}{2}} \cos^n x\, dx = \frac{(n-1)}{n} \frac{(n-3)}{(n-2)} \frac{(n-5)}{(n-4)} \cdots\cdots\cdots \frac{6}{7} \frac{4}{5} \frac{2}{3}$

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Question No : 2 of 52 Marks: 1 (Budgeted Time 1 Min)

What is the amplitude of a periodic function defined by $f(x) = 4\cos 3x$?

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Answer (Please select your correct option)

- 1
- 3
- 4 correct
- 12

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Question No : 3 of 52 Marks: 1 (Budgeted Time 1 Min)

What is the period of a periodic function defined by $f(x) = \sin\frac{x}{2}$?

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Answer (Please select your correct option)

- $\frac{\pi}{2}$
- π correct
- $\frac{3\pi}{2}$
- 4π

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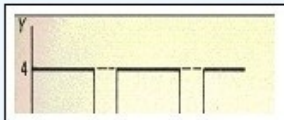
Question No : 4 of 52 **Marks: 1 (Budgeted Time 1 Min)**

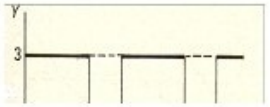
Match the following periodic function with its graph.

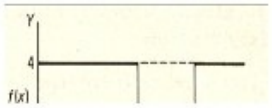
$$f(x) = \begin{cases} 4 & 0 < x < 5 \\ 0 & 5 < x < 8 \end{cases}$$

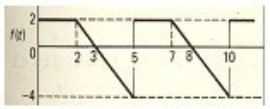
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Answer (Please select your correct option)









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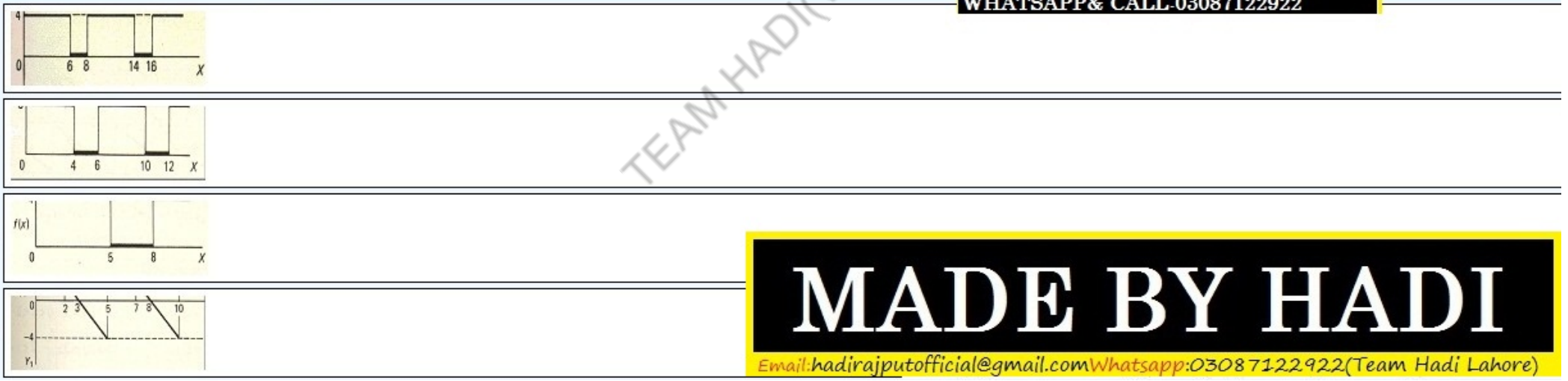
Question No : 4 of 52 **Marks: 1 (Budgeted Time 1 Min)**

Match the following periodic function with its graph.

$$f(x) = \begin{cases} 4 & 0 < x < 5 \\ 0 & 5 < x < 8 \end{cases}$$

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Answer (Please select your correct option)



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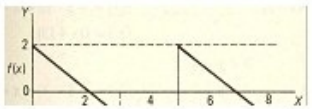
Question No : 5 of 52 Marks: 1 (Budgeted Time 1 Min)

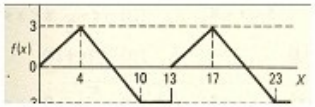
Match the following periodic function with its graph.

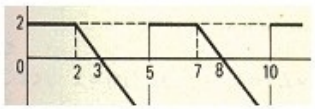
$$f(x)=\begin{cases}2-x & 0<x<3\\ -1 & 3<x<5\end{cases}$$

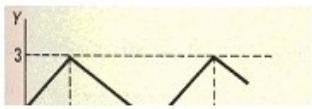
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Answer (Please select your correct option)









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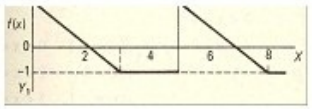
Question No : 5 of 52 **Marks: 1 (Budgeted Time 1 Min)**

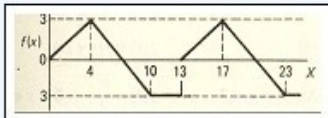
Match the following periodic function with its graph.

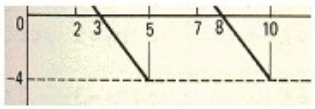
$$f(x)=\begin{cases}2-x & 0<x<3\\ -1 & 3<x<5\end{cases}$$

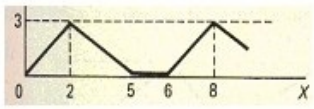
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Answer (Please select your correct option)









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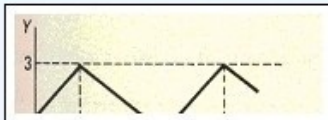
Question No : 6 of 52 Marks: 1 (Budgeted Time 1 Min)

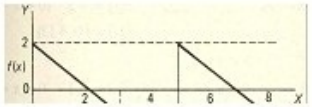
Match the following periodic function with its graph.

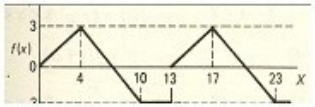
$$f(x) = \begin{cases} \frac{3}{4}x & 0 < x < 4 \\ 7 - x & 4 < x < 10 \\ -3 & 10 < x < 13 \end{cases}$$

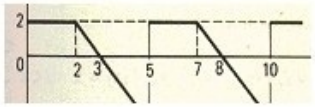
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Answer (Please select your correct option)









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correct

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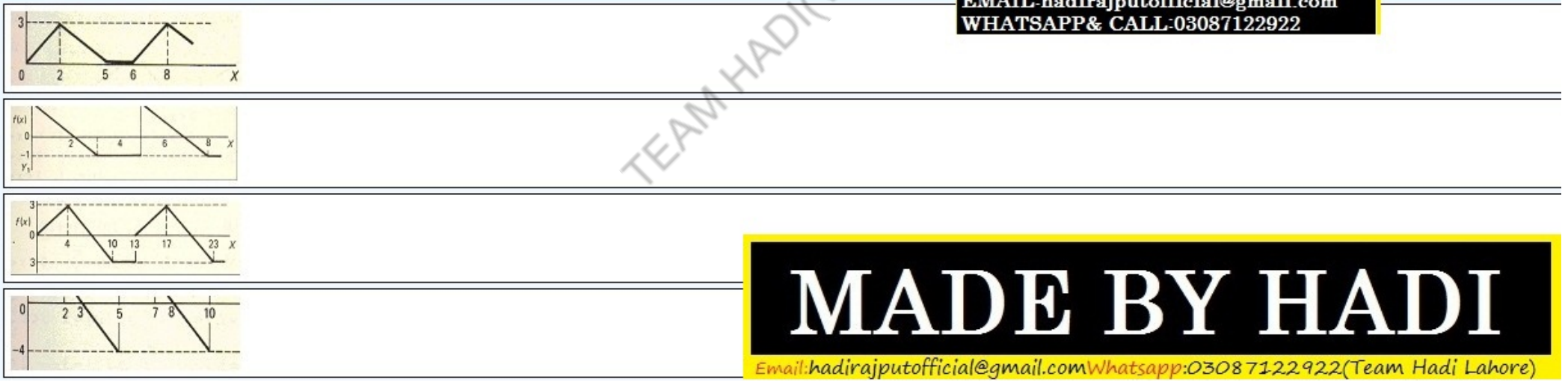
Question No : 6 of 52 **Marks: 1 (Budgeted Time 1 Min)**

Match the following periodic function with its graph.

$$f(x) = \begin{cases} \frac{3}{4}x & 0 < x < 4 \\ 7 - x & 4 < x < 10 \\ -3 & 10 < x < 13 \end{cases}$$

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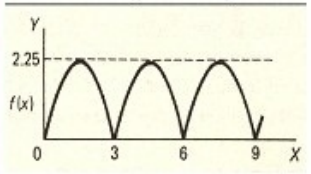
Answer (Please select your correct option)



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Question No : 7 of 52 Marks: 1 (Budgeted Time 1 Min)

What is the period of periodic function whose graph is as below?



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Answer (Please select your correct option)

- 0
- 2.25
- 3 correct
- 6

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Question No : 8 of 52 Marks: 1 (Budgeted Time 1 Min)

The function $f(x) = x^3 e^x$ is ----------

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Answer (Please select your correct option)

- Even function
- Odd function
- Neither even nor odd

correct

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Question No : 9 of 52 Marks: 1 (Budgeted Time 1 Min)

The graph of an even function is symmetrical about ----------

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Answer (Please select your correct option)

- x-axis
- y-axis

correct

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- origin

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Question No : 10 of 52 Marks: 1 (Budgeted Time 1 Min)

Which of the following is Laplace inverse transform of the function $f(s)$ defined by $f(s) = \frac{3}{s-2} - \frac{2}{s}$?

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Answer (Please select your correct option)

- $3te^{2t} - 2$
- $3e^{2t} - 2t$
- $3e^{2t} - 2$
- None of these.

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Question No : 11 of 52 Marks: 1 (Budgeted Time 1 Min)

What is the value of $L\{e^{5t}\}$ if L denotes laplace transform?

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Answer (Please select your correct option)

- $L\{e^{5t}\} = \frac{1}{s-5}$ correct
- $L\{e^{5t}\} = \frac{s}{s^2+25}$
- $L\{e^{5t}\} = \frac{5}{s^2+25}$
- $L\{e^{5t}\} = \frac{5!}{s^6}$

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Question No : 12 of 52 Marks: 1 (Budgeted Time 1 Min)

What is laplace transform of the function $F(t)$ if $F(t) = \sin 3t$?

Answer (Please select your correct option)

- $L\{\sin 3t\} = \frac{3}{s^2+9}$ correct
- $L\{\sin 3t\} = \frac{s}{s^2+9}$
- $L\{\sin 3t\} = \frac{1}{s-3}$
- $L\{\sin 3t\} = \frac{3!}{s^4}$

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Question No : 13 of 52 Marks: 1 (Budgeted Time 1 Min)

If L denotes laplace transform then
$L(te^{5t})$ =

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Answer (Please select your correct option)

- $L(te^{5t}) = \frac{1}{s^2 - 5}$
- $L(te^{5t}) = \frac{1}{s^2 + 5}$
- $L(te^{5t}) = \frac{1}{(s+5)^2}$
- $L(te^{5t}) = \frac{1}{(s-5)^2}$ correct

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Question No : 14 of 52 Marks: 1 (Budgeted Time 1 Min)

What is Laplace Inverse Transform of $\frac{s}{s^2+25}$

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Answer (Please select your correct option)

- $L^{-1}\left\{\frac{s}{s^2+25}\right\} = \sin 5t$
- $L^{-1}\left\{\frac{s}{s^2+25}\right\} = \cos 5t$
- $L^{-1}\left\{\frac{s}{s^2+25}\right\} = \sin 25t$
- $L^{-1}\left\{\frac{s}{s^2+25}\right\} = \cos 25t$

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Question No : 15 of 52 Marks: 1 (Budgeted Time 1 Min)

Divergence of a vector function is always a ----------

Answer (Please select your correct option)

- Scalar

correct

- Vector

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Question No : 16 of 52 Marks: 1 (Budgeted Time 1 Min)

If p is the period of a function then that function is said to be periodic if $f(x+p) = f(x)$, _____

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Answer (Please select your correct option)

- For all values of x in the domain of f
- For positive values of x in the domain of f
- For negative values of x in the domain of f

correct

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Question No : 17 of 52 Marks: 1 (Budgeted Time 1 Min)

Which of the following is associated to each point on a plane?

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Answer (Please select your correct option)

- A real number

correct

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- A natural number
- An ordered pair

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Question No : 18 of 52 Marks: 1 (Budgeted Time 1 Min)

Consider the interval (-1, 5) on co-ordinate line. What does this interval denote?

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Answer (Please select your correct option)

- The set of all real numbers between -1 and 5 correct
- The set of all integers between -1 and 5
- The set of all natural numbers between -1 and 5
- The set of all rational numbers between -1 and 5

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Question No : 19 of 52 Marks: 1 (Budgeted Time 1 Min)

Equation of the circular disk with radius a and origin at $(0, 0)$ is given by . . . . . .

Answer (Please select your correct option)

- $x^2 + y^2 = a^2$ correct
- $x^2 + y^2 \geq a^2$
- $x^2 + y^2 \leq a^2$

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Question No : 20 of 52 Marks: 1 (Budgeted Time 1 Min)

In three dimensional space, the equation $y = x^2$ always represents . . . . . .

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Answer (Please select your correct option)

- Parabola

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- Straight line
- Half cylinder
- Cone

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Question No : 21 of 52 Marks: 1 (Budgeted Time 1 Min)

Domain of the function $f(x, y, z) = \sqrt{x^2 + y^2 + z^2}$ is . . . . . .

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Answer (Please select your correct option)

- Entire 3D-Space

correct

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- Entire 3D-Space except origin
- First octant of 3D-Space

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Question No : 22 of 52 Marks: 1 (Budgeted Time 1 Min)

Suppose $f(x, y) = x^3e^{xy}$. Which of the following options is correct?

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Answer (Please select your correct option)

- $\frac{\partial f}{\partial y} = 3x^3e^{xy}$
- $\frac{\partial f}{\partial y} = x^3e^{xy}$
- $\frac{\partial f}{\partial y} = x^4e^{xy}$
- $\frac{\partial f}{\partial y} = x^3ye^{xy}$ ⇦

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Question No : 23 of 52 Marks: 1 (Budgeted Time 1 Min)

Gradient of a scalar function always results in a . . . . . function.

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Answer (Please select your correct option)

- Scalar
- Continuous
- Vector
- Constant

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Question No : 24 of 52 Marks: 1 (Budgeted Time 1 Min)

Let x, y, z be the length, width and height of a rectangular box. The area of bottom will be . . . . . .

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Answer (Please select your correct option)

- $A = yz$
- $A = xz$
- $A = xy$
- $A = xyz$

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Question No : 25 of 52 Marks: 1 (Budgeted Time 1 Min)

Let x, y, z be the length, width and height of an open rectangular box. The surface area of the box will be . . . . . .

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Answer (Please select your correct option)

- $A = xy + 2yz + 2xz$
- $A = yz + 4$
- $A = xz + yz + zx$
- $A = xyz$ ⇦

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Question No : 26 of 52 Marks: 1 (Budgeted Time 1 Min)

If $R = \{(x, y) : 0 \le x \le 2 \text{ and } 0 \le y \le 3\}$, then $\iint_R (1 - ye^{xy}) dA = \ldots\ldots\ldots$

Answer (Please select your correct option)

- $\int_0^2 \int_0^3 (1 - ye^{xy}) dy dx$ correct
- $\int_0^2 \int_0^3 (1 - ye^{xy}) dx dy$
- $\int_2^3 \int_0^0 (1 - ye^{xy}) dx dy$
- $\int_0^2 \int_2^3 (4xe^{2y}) dy dx$

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Question No : 27 of 52 Marks: 1 (Budgeted Time 1 Min)

Match the following equation in polar co-ordinates with its graph.

$r \sin\theta = a$

where a is an arbitrary constant

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Answer (Please select your correct option)



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Question No : 27 of 52 Marks: 1 (Budgeted Time 1 Min)

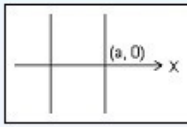
Match the following equation in polar co-ordinates with its graph.

$r \sin\theta = a$

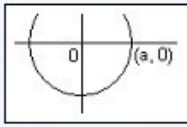
where a is an arbitrary constant

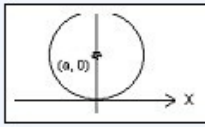
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Answer (Please select your correct option)









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Question No : 28 of 52 Marks: 1 (Budgeted Time 1 Min)

Polar co-ordinates of a point are $(-8, 45^0)$. Which of the following is another possible polar co-ordinates representation of this point?

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Answer (Please select your correct option)

- $(8, 225^0)$
- $(-8, 225^0)$
- $(8, 45^0)$
- $(8, 315^0)$

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Question No : 29 of 52 Marks: 1 (Budgeted Time 1 Min)

Polar co-ordinates of a point are $\left(7, \frac{-\pi}{4}\right)$. Which of the following is another possible polar co-ordinates representation of this point?

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Answer (Please select your correct option)

- $\left(7, \frac{3\pi}{4}\right)$
- $\left(-7, \frac{3\pi}{4}\right)$
- $\left(-7, \frac{-\pi}{4}\right)$
- $\left(7, \frac{-3\pi}{4}\right)$

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Question No : 30 of 52 Marks: 1 (Budgeted Time 1 Min)

If $p(r,\theta)$ is a point in polar coordinate system, then r is the distance of p from . . . . . .

Answer (Please select your correct option)

- Pole
- Imaginary axis
- None of these
- Polar axis

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Question No : 31 of 52 Marks: 1 (Budgeted Time 1 Min)

Given the integral $\iint_R f(x,y)dxdy$, after converting to polar coordinates, it will become . . . . . where $a \le \theta \le b$ and c ≤ r ≤ d.

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Answer (Please select your correct option)

- $\int_a^b \int_c^d f(r,\theta)rdrd\theta$ correct
- $\int_a^b \int_c^d f(r,\theta)drd\theta$
- $\int_a^b \int_c^d f(r,\theta)rd\theta dr$
- $\int_a^b \int_c^d f(r,\theta)d\theta dr$

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Question No : 32 of 52 Marks: 1 (Budgeted Time 1 Min)

The graph of the equation $r = 3\hat{i} - 2\hat{j} - \hat{k}$ is the point . . . . . .

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Answer (Please select your correct option)

- (1, −2, 3)
- (3, −2, −1) correct
- (3, 2, 1)
- (−3, −2, −1)

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Question No : 33 of 52 Marks: 1 (Budgeted Time 1 Min)

The natural domain of a vector valued function is the . . . . . of the natural domains of its components

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Answer (Please select your correct option)

- Product
- Summation
- Union
- Intersection

correct

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Question No : 34 of 52 Marks: 1 (Budgeted Time 1 Min)

If $\vec{R}(t)$ is the anti-derivative of a given vector valued function $\vec{r}(t)$ i.e., $\vec{R}'(t) = \vec{r}(t)$ then $\int_a^b \vec{r}(t)dt = \ldots\ldots\ldots$

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Answer (Please select your correct option)

- $\vec{R}(a) + \vec{R}(b)$
- $\vec{R}(b) - \vec{R}(a) + c$ where c is a non-zero constant.
- $\vec{R}(a).\vec{R}(b)$
- $\vec{R}(b) - \vec{R}(a)$

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Question No : 35 of 52 Marks: 1 (Budgeted Time 1 Min)

For the given vector valued functions $\vec{r}_1(t)$ and $\vec{r}_2(t)$, $\frac{d}{dt}[\vec{r}_1(t)\times\vec{r}_2(t)]$ =..... where "×" and "·" represent the cross and dot product respectively.

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Answer (Please select your correct option)

- $\vec{r}_1 \cdot \frac{d\vec{r}_2}{dt} + \frac{d\vec{r}_1}{dt} \cdot \vec{r}_2$
- $\vec{r}_1 \times \frac{d\vec{r}_2}{dt} + \frac{d\vec{r}_1}{dt} \times \vec{r}_2$
- $\frac{d\vec{r}_2}{dt} \times \vec{r}_1 + \vec{r}_2 \times \frac{d\vec{r}_1}{dt}$
- $\frac{d\vec{r}_2}{dt} \times \vec{r}_1 + \frac{d\vec{r}_1}{dt} \times \vec{r}_2$

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Question No : 36 of 52 Marks: 1 (Budgeted Time 1 Min)

A vector valued function $\vec{r}(t) = x(t)\hat{i} + y(t)\hat{j} + z(t)\hat{k}$ is continuous if . . . . . .

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Answer (Please select your correct option)

- Atleast one of its components is continuous.
- All of its components are necessarily differentiable.
- All of its components are continuous.
- Limit exists for all of its components.

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Question No : 37 of 52 Marks: 1 (Budgeted Time 1 Min)

Given a vector valued function $\vec{r}(t) = \frac{1}{(t-3)}\hat{i} + e^t\hat{j}$ and its anti-derivative $\vec{R}(t) = \ln(t-3)\hat{i} + e^t\hat{j}$, then $\int \vec{r}(t)dt = \ldots\ldots\ldots\ldots$

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Answer (Please select your correct option)

- $\ln(t-3)\hat{i} + e^t\hat{j} + c$
- $(t-3)\hat{i} + \frac{e^t}{2}\hat{j} + c$
- $(t-3)^{-1}\hat{i} + \frac{e^t}{2}\hat{j} + c$
- $\frac{1}{(t-3)}\hat{i} + e^t\hat{j}$ correct

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Question No : 38 of 52 Marks: 1 (Budgeted Time 1 Min)

For any two vector valued functions $\vec{r}_1(t)$ and $\vec{r}_2(t)$, $\frac{d}{dt}[\vec{r}_1(t).\vec{r}_2(t)] =$ ..... where "×" and "·" represent the cross and dot product respectively.

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Answer (Please select your correct option)

- $\vec{r}_1 \cdot \frac{d\vec{r}_2}{dt} - \frac{d\vec{r}_1}{dt} \cdot \vec{r}_2$
- $\vec{r}_1 \cdot \frac{d\vec{r}_2}{dt} + \frac{d\vec{r}_1}{dt} \cdot \vec{r}_2$
- $\vec{r}_1 \times \frac{d\vec{r}_2}{dt} + \frac{d\vec{r}_1}{dt} \times \vec{r}_2$
- $\vec{r}_1 \cdot \frac{d\vec{r}_2}{dt} - \frac{d\vec{r}_1}{dt} \times \vec{r}_2$ correct

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Question No : 39 of 52 Marks: 1 (Budgeted Time 1 Min)

A single curve can be represented by . . . . . vector valued function(s).

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Answer (Please select your correct option)

- Two
- Infinitely many
- Single

correct

- Three

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Question No : 40 of 52 Marks: 1 (Budgeted Time 1 Min)

A function is said to be smooth if it's derivative is . . . . . on any value of its domain.

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Answer (Please select your correct option)

- continuous and non zero
- piecewise continuous
- defined and non zero
- differentiable

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Question No : 41 of 52 Marks: 2 (Budgeted Time 4 Min)

Use Wallis sine formula to evaluate $\int_0^{\frac{\pi}{2}} \sin^5 x \, dx$

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Answer (Please click here to Add Answer)

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Question No : 42 of 52 Marks: 2 (Budgeted Time 4 Min)

Find Laplace transform of the function $F(t)$ if $F(t) = \cos 3t$.

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Answer (Please click here to Add Answer)

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Question No : 43 of 52 Marks: 2 (Budgeted Time 4 Min)

State the condition when $\iint_R f(x,y)dA = \int_c^d \int_a^b f(x,y)dxdy = \int_a^b \int_c^d f(x,y)dydx$ where R is the region of integration.

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Answer (Please click here to Add Answer)

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Question No : 44 of 52 Marks: 2 (Budgeted Time 4 Min)

Find derivative of the following vector-valued function.

$$\vec{r}(t) = e^{t^2}\,\hat{i} + t^2\,\hat{j} + \sec 2t\,\hat{k}$$

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Answer (Please click here to Add Answer)

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Question No : 45 of 52 Marks: 3 (Budgeted Time 6 Min)

Use Wallis sine formula to evaluate $\int_0^{\frac{\pi}{2}} \left(\sin^3 x + \sin^4 x\right) dx$

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Answer (Please click here to Add Answer)

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Question No : 46 of 52 Marks: 3 (Budgeted Time 6 Min)

Find Laplace transform of the function $F(t)$ if

$F(t) = e^{2t} \sin 3t$

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Answer (Please click here to Add Answer)

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Question No : 47 of 52 Marks: 3 (Budgeted Time 6 Min)

Find the critical point for the given function $f(x, y) = 6x^2 + xy - 2y^2$ along the line $y = 3x + 1$ at which the absolute extrema of the function can occur.

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Answer (Please click here to Add Answer)

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Question No : 48 of 52 Marks: 3 (Budgeted Time 6 Min)

What is the arc-length of the curve $\vec{r}(t) = 3\cos t\,\hat{i} + 3\sin t\,\hat{j}$ when $0 \le t \le 2\pi$?

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Answer (Please click here to Add Answer)

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Question No : 49 of 52 Marks: 5 (Budgeted Time 10 Min)

Determine the Fourier co-efficient b_n of the periodic function defined below:

$f(x) = 2x + 1 \qquad 0 < x < 2$

Answer (Please click here to Add Answer)

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Question No : 50 of 52 Marks: 5 (Budgeted Time 10 Min)

Determine whether the following vector field $\vec{F}$ is conservative or not.

$\vec{F}(x, y, z) = x^2 z\,\hat{i} + y^2 x\,\hat{j} + (y + 2z)\,\hat{k}$

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Answer (Please click here to Add Answer)

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Question No : 51 of 52 Marks: 5 (Budgeted Time 10 Min)

If $f(x, y) = (x - 4)\ \ln(xy)$

Find both first order partial derivatives.

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Answer (Please click here to Add Answer)

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Question No : 52 of 52 Marks: 5 (Budgeted Time 10 Min)

Let $\vec{r}(t) = t^2\,\hat{i} + t\,\hat{j} + (t^2 - 5)\,\hat{k}$. Find t, such that $\vec{r}(t)$ and $\vec{r}'(t)$ are perpendicular to each other.

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Answer (Please click here to Add Answer)

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